Rules of Inference

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1. Modus Ponens (Law of Detachment)
   If a conditional statement is true and its condition is satisfied, the conclusion must be true.
   $$\text{If } p \to q \text{ and } p, \text{ then } q$$
2. Modus Tollens (Law of Contrapositive)
   If a conditional statement is true, and its consequent is false, then its antecedent must also be false.
   $$\text{If } p \to q \text{ and } \neg q, \text{ then } \neg p$$
3. Hypothetical Syllogism
   If two conditional statements are true, where the consequent of the first is the antecedent of the second, then a third conditional statement combining the antecedent of the first and the consequent of the second is also true.
   $$\text{If } p \to q \text{ and } q \to r, \text{ then } p \to r$$
4. Disjunctive Syllogism If a disjunction is true, and of its parts is false, then the other part must be true.
   $$\text{If } p \vee q \text{ and } \neg p \text{, then } q$$
5. Conjunction
   If two statements are true, then their conjunction is also true.
   $$\text{If } p \text{ and } q, \text{ then } p \wedge q$$
6. Simplification
   If a conjunction is true, then each of its conjuncts are also true.
   $$\text{If } p \wedge q, \text{ then } p$$
7. Addition
   If a statement is true, then the disjunction of that statement with any other statement is also true.
   $$\text{If } p, \text{ then } p \vee q$$
8. Absorption
   If a conditional statement is true, then the antecedent implies a conjunction of itself and the consequent.
   $$\text{If } p \to q, \text{ then } p \to (p \wedge q)$$
9. Resolution
   If two disjunctions are true, and one containing proposition p while the other contains its negation $\neg p$, then the disjunction of the remaining parts is true.
   $$\text{If } p \vee q \text{ and } \neg p \vee r, \text{ then } q \vee r$$